

CONCISE DESCRIPTION OF RELEVANCE: THIRD-PARTY SUBMISSION UNDER 37 CFR § 1.290

U.S. Application No. 19/318,450 (Publication No. US 2026/0070232 A1)

Document 2: WO 2010135788 A1 (“Document 2”)

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Title: Switchable magnetic implement

This document discloses a switchable magnetic implement for securing workpieces, which can be used as a magnetic gripper, featuring movable magnets and pole extension members with optimized engagement surfaces for improved flux transfer. Its potential relevance to the claims of the above-identified application is set forth below.

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 1: “A magnetic gripper for gripping a ferromagnetic workpiece, said magnetic gripper comprising”	The reference describes a switchable magnetic implement for securing work pieces for and during metal working applications, which is a magnetic gripper for gripping a ferromagnetic workpiece. “The present invention relates to switchable magnetic devices used to secure work pieces for and during metal working applications (such as welding, machining, joining, etc) or to serve as<'>holders for ancillary tools used during such metal working or generally as ...” • Paragraph [0001]
Claim 1: “a magnet which is arranged in the housing and which is displaceable along a displacement direction between a gripping position for gripping the ferromagnetic workpiece and a release position for releasing the ferromagnetic workpiece”	The reference discloses that both magnets 11, 12 are disc or cylindrically shaped magnets which are diametrically magnetised, and are movable to selectively generate an external magnetic field. “Both magnets 11 , 12 are disc or cylindrically shaped magnets which are diametrically magnetised, Ie the north pole and the south pole of the magnets are located on opposite sides of a vertical plane that extends over the hight ...” • Paragraph [0001]
Claim 2 (The magnetic gripper according to claim 1, wherein the housing extends along a longitudinal axis)	The reference discloses that both magnets 11, 12 are disc or cylindrically shaped magnets, implying the housing extends along a longitudinal axis. “Both magnets 11 , 12 are disc or cylindrically shaped magnets which are diametrically magnetised, Ie the north pole and the south pole of the magnets are located on opposite sides of a vertical plane that extends over the hight ...” • Paragraph [0001]

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Claim 3 (The magnetic gripper according to claim 1, wherein each pole shoe has an active structure for directing the magnetic field portion)	The reference discloses that the contact or engagement surfaces of the pole extension members are optimised to provide an improved flux transfer functionality, thereby directing the magnetic field. “Relevant is that the contact or engagement surfaces that provide the interface between pole extension members and a work piece, are optimised in order to provide an improved flux transfer functionality between the active magnetic material (permanent magnets) through the ...” • Paragraph [0005]
Claim 4 (The magnetic gripper according to claim 3, wherein the active structure has a bevel)	The reference discloses a pole extension member 130 that has a lower engagement edge surface which is 'pitched' or 'notched'. “Figure 2c illustrates a second embodiment of a pole extension member 130, which is again a ferromagnetic steel plate 132' with two long holes 134 extending through the plate 132'. However, instead of having a straight lower engagement edge surface ...” • Paragraph [0012]
Claim 5 (The magnetic gripper according to claim 3, wherein the active structure has a plurality of projections for directing the magnetic field portion and for forming the workpiece ...)	The reference discloses multi-piece extension members 150 and 160, where pole extension member 150 consists of a plurality of thin, stacked ferromagnetic plates 151, which can be considered projections for forming the workpiece contact surface. “In another embodiment of the modularly attachable pole extension members, figures 3a and 3b illustrate multi-piece extension members 150 and 160, respectively. In figure 3a, the pole extension member 150 consists of a plurality of thin, stacked ferromagnetic plates 151 ...” • Paragraph [0013]
Claim 8 (The magnetic gripper according to claim 1, wherein the number of pole shoes includes a first pole shoe and a second pole shoe)	The reference discloses that the magnetic device 10 essentially consists of two permanent magnets 11, 12 located within a housing 15 consisting of two pole pieces 16, 17, which function as pole shoes. “The magnetic device 10 essentially consist of two permanent magnets 11 , 12 located within a housing 15 consisting of two pole pieces 16, 17 ...” • Paragraph [0001]

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 13 (The magnetic gripper according to claim 1, wherein a housing portion of the housing is designed to largely suppress the escape of the magnetic field of the magnet from the housing ...)	The reference discloses that in the described arrangement, the first and second magnets act as an internal active magnetic shunt, resulting in a quite low external magnetic field strength from the device, thereby largely suppressing the escape of the magnetic field. “In this arrangement, the first and second magnets act as an internal active magnetic shunt and as a result the external magnetic field strength from the device is quite low (and ideally not present).” • Paragraph [0002]
Claim 14 (A magnetic gripper system, the magnetic gripper system comprising)	The reference describes switchable magnetic devices used to secure work pieces for and during metal working applications, which constitutes a magnetic gripper system. “The present invention relates to switchable magnetic devices used to secure work pieces for and during metal working applications (such as welding, machining, joining, etc) or to serve as<'>holders for ancillary tools used during such metal working or generally as ...” • Paragraph [0001] • Paragraph [0002]